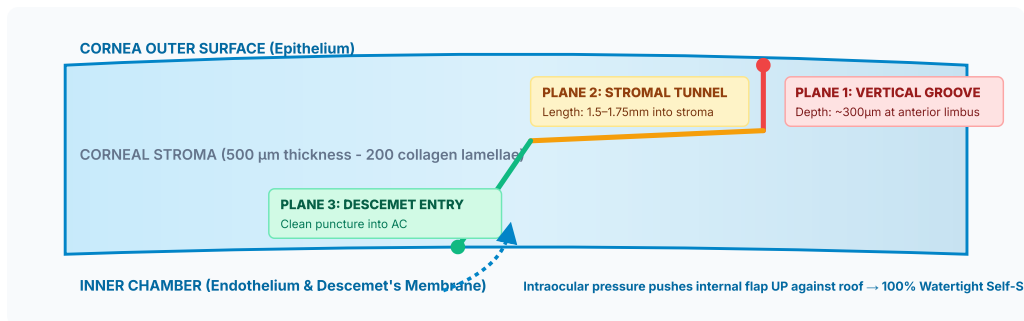


Comprehensive Ophthalmic Surgical Field Manual

Phacoemulsification, Foldable IOL Implantation & Nd:YAG Laser Capsulotomy

A complete, beginner-accessible clinical compendium detailing step-by-step anterior segment surgery, microscopic corneal incision architecture, fluidic dynamics, and complications management.



OFFICIAL SURGICAL SIMULATOR PROTOCOL & TRAINING COMPENDIUM

Standardized for Ophthalmic Residents, Fellows & Surgical Technicians • 2026 Edition

Surgical Modules & Clinical Curriculum

Module 1: Cataract Phacoemulsification Surgery

Paracentesis, Tri-Planar Wound, OVD Cushion, CCC, Hydrodissection, Phaco-Chop & I/A

MODULE 1**Module 2: Foldable Intraocular Lens (IOL) Implantation**

Bag Refill, Cartridge Delivery, Leading Haptic, Sinskey Hook Centering & OVD Washout

MODULE 2**Module 3: Nd:YAG Laser Posterior Capsulotomy**

Abraham Lens, HeNe Aiming Alignment, +150µm Posterior Defocus & Cruciate Laser Breakdown

MODULE 3**Module 4: Critical Surgical Anatomy & Fluidics Reference Guide**

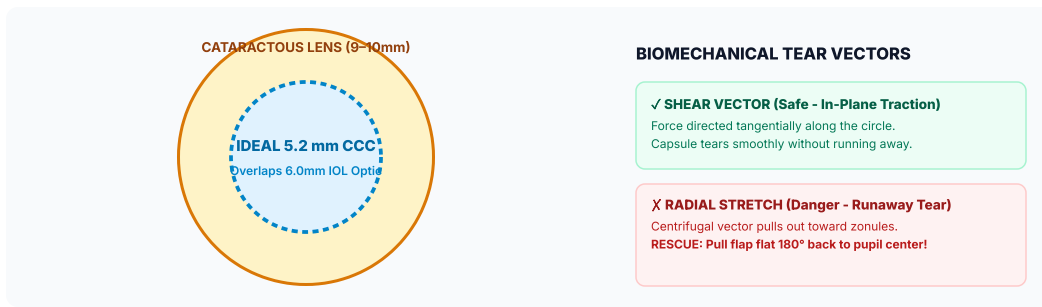
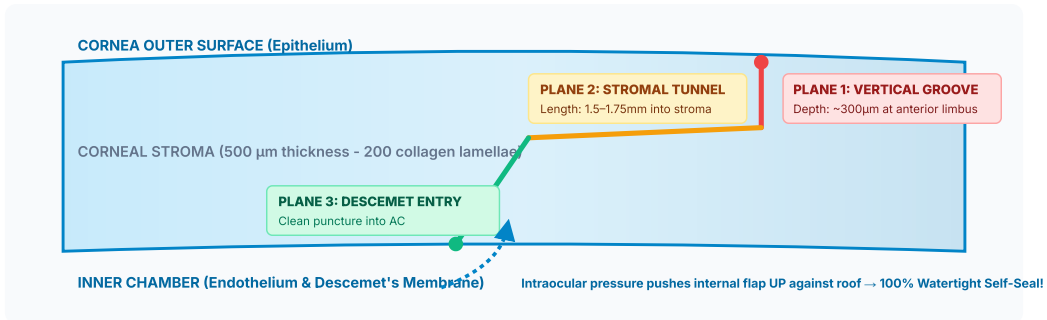
Corneal Endothelium Preservation, Post-Occlusion Surge Equations & CDE Calculation

MODULE 4**How To Use This Manual**

Each module is written specifically so that someone with zero previous ophthalmic experience can understand both **what physical action to take** and **why that step is vital to preserving the patient's vision**. Key safety alerts, anatomical tissue planes, instrument dimensions, and complication avoidance rules are provided for each phase of surgery.

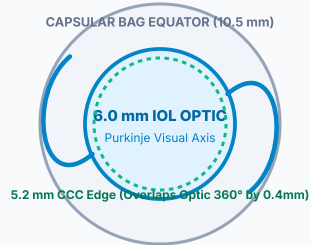
Cataract Phacoemulsification Protocol

Surgical Objective: Clear the clouded crystalline lens and evacuate cortical debris with minimal Cumulative Dissipated Energy (CDE < 15 %-sec) and near-zero endothelial cell loss.



Foldable IOL Implantation Protocol

Surgical Objective: Deliver a foldable 6.0mm acrylic optic safely into the capsular bag, establish 360° anterior capsular overlap, and evacuate all retro-lens viscoelastic to prevent acute IOP elevation.



WHY 360° OVERLAP MATTERS

1. Prevents PCO (Secondary Cataracts)

Square-edge optic + shrink-wrap overlap traps residual lens cells, stopping cloudy migration.

2. Perfect Centration & No Lens Tilt

Both spring arms seated symmetrically in bag prevents unwanted astigmatism and double vision.

Nd:YAG Laser Capsulotomy Protocol

Surgical Objective: Restore optical axis clarity through photodisruption of the opacified posterior capsule using cruciate laser cuts with +150 μ m posterior offset to prevent IOL pitting.

